

Supply Chain Analytics for Nassau Candy Distributor

*Exploratory Data Analysis, Operational Insights,
and Factory Reassignment Recommendations*

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Abstract

This paper presents a comprehensive data-driven analysis of the Nassau Candy Distributor supply chain, utilising **10,194 order records** spanning 2024–2025 across the United States and Canada. The dataset encompasses 15 confectionery products across three divisions (Chocolate, Sugar, and Other), fulfilled by five geographically distributed factories serving four sales regions. Following data cleaning and feature engineering – including lead time derivation, factory mapping, and haversine-based shipping distance computation – we conduct multi-dimensional exploratory analysis of lead time distribution, factory and regional performance, product-level profitability, and route efficiency. K-Means clustering ($K = 5$, silhouette-validated) identifies five operationally distinct route categories, ranging from Optimised to Bottleneck, while the product clustering ($K = 3$) stratifies all 15 SKUs by delivery speed and margin profile. Predictive modelling via Linear Regression, Random Forest, and Gradient Boosting Regressors – evaluated on RMSE, MAE, and R^2 – selects Linear Regression as the best-performing model ($R^2 = 0.53$), confirming that order year is the dominant driver of lead time ($r = 0.72$) and that operational variables contribute minimally under the current static assignment structure. A scenario simulation engine subsequently evaluates **60 reassignment scenarios** across all 13 active products, identifying **The Other Factory** as the universally optimal reassignment destination, projecting **534,218 cumulative operational days saved** at an average lead-time reduction of **54.8 days per product** with negligible profit-margin impact ($\Delta < 0.05\%$). These findings expose a critical underutilisation of The Other Factory – currently handling under 1% of total order volume – and provide actionable, quantified recommendations for supply-chain restructuring that improves shipping efficiency without sacrificing profitability.

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1. Introduction

Efficient supply-chain management is among the most consequential operational challenges for consumer-goods distributors. Delays in order fulfilment — measured as *lead time*, the interval in days between order placement and shipment — directly translate into customer dissatisfaction, elevated holding costs, and forgone revenue. For candy and confectionery distributors operating across seasonal demand cycles and varied regional markets, optimising factory-to-region routing is especially critical to maintaining both service levels and profitability.

Nassau Candy Distributor operates a multi-factory, multi-region distribution network spanning the continental United States and Canada. Products are sourced from five manufacturing facilities: *Lot's O' Nuts*, *Wicked Choccy's*, *Secret Factory*, *The Other Factory*, and *Sugar Shack*. Orders are routed to four geographic sales regions: Atlantic, Gulf, Interior, and Pacific. While the network handles over 10,000 annual orders across 15 SKUs in three product divisions, factory-to-product assignments are governed by static legacy rules rather than data-driven optimisation. This creates measurable inefficiencies: suboptimal shipping distances, high lead times in specific region-product combinations, and margin pressure from avoidable logistics costs.

This project addresses those gaps by introducing a decision-intelligence framework that moves Nassau Candy from descriptive reporting to actionable operational recommendations. Specifically, the objectives of this analysis are:

- i. To characterise the structure and quality of the order dataset through exploratory data analysis (EDA);
- ii. To uncover lead-time patterns across factories, regions, ship modes, and products;
- iii. To quantify financial performance — sales, cost, and profit margin — at the factory and product level;
- iv. To cluster factory-region routes and individual products based on operational efficiency and financial profile;
- v. To train and evaluate machine-learning models predicting order lead time, with selection based on RMSE, MAE, and R^2 ;
- vi. To simulate factory reassignment scenarios and rank recommendations by a weighted composite priority score balancing speed, profitability, and order volume.

The analysis is structured to mirror the decision pipeline a supply-chain analyst would follow: understand the data, identify patterns, model outcomes, and recommend action. The remainder of the paper proceeds as follows: Section 2 reviews relevant prior work;

Section 3 describes the dataset and preprocessing methodology; Section 4 presents the exploratory analysis; Section 5 synthesises cross-dimensional operational findings; Section 6 covers predictive modelling and feature importance; Section 7 details route and product clustering; Section 8 presents the factory reassignment simulation and ranked recommendations; and Section 9 concludes with limitations and directions for future work.

2. Related Work

Supply chain network design and factory-to-region assignment fall within a well-established body of operations research literature. Foundational work by [Dantzig and Ramser \(1959\)](#) on the vehicle routing problem established the combinatorial structure underpinning factory assignment optimisation, demonstrating that even modest changes in routing rules can yield material reductions in total logistics cost. [Chopra and Meindl \(2007\)](#) subsequently showed that strategic network-level decisions — including facility location and product allocation across manufacturing sites — account for a disproportionate share of total supply chain expenditure, motivating the simulation-based reassignment framework adopted in this study rather than a purely operational scheduling approach. More recent survey work on distribution network design ([Farahani et al., 2011](#)) has reinforced the importance of quantifying the operational impact of reassignment *before* execution, a gap this project directly addresses through its scenario simulation engine.

Cluster analysis has emerged as a widely used technique for segmenting logistics networks by performance profile. [MacQueen \(1967\)](#) introduced the K-Means algorithm, which partitions observations into K clusters by minimising within-cluster sum of squared distances. The Silhouette coefficient, proposed by [Rousseeuw \(1987\)](#), provides an internal validation metric for selecting the optimal K without requiring ground-truth labels — the method applied here to validate $K = 5$ for route clustering and $K = 3$ for product clustering. Applications of K-Means in logistics contexts include route segmentation for freight optimisation, warehouse zone classification, and demand-region grouping for inventory planning, where the algorithm’s interpretability supports managerial decision-making alongside its computational tractability on industrial-scale datasets.

Predictive modelling of lead time has attracted growing attention as organisations seek to move from reactive to proactive logistics management. [Breiman \(2001\)](#) introduced Random Forests as a variance-reducing ensemble method, while [Friedman \(2001\)](#) developed Gradient Boosting Machines as a sequential alternative capable of capturing complex non-linear feature interactions. Both methods have been applied to order lead-time forecasting in manufacturing and e-commerce settings, typically yielding R^2 values of 0.5 to 0.8 depending on dataset richness and feature diversity. The finding in this study that Linear Regression matches or exceeds both ensemble methods ($R^2 = 0.53$) is consistent

with prior evidence that ensemble complexity provides no marginal benefit when a single dominant feature accounts for the overwhelming majority of predictive signal ([Breiman, 2001](#)). Finally, the Haversine formula ([Sinnott, 1984](#)) — used here to compute great-circle shipping distances between factory coordinates and regional centroids — is a standard tool in geographic information systems and logistics for estimating straight-line travel distance over a spherical Earth surface.

3. Data and Methodology

3.1 Dataset Description

The primary dataset, `Nassau Candy Distributor.csv`, contains **10,194 rows** and **18 columns** prior to feature engineering. Each row represents one order line. The dataset was provided as part of a data science internship programme. Table 1 summarises the original schema, data types, and representative values.

Table 1: Dataset Schema — Original 18 Columns

Column	Dtype	Description / Representative Values
Row ID	int64	Unique row index (1–10,194)
Order ID	object	Alphanumeric order identifier (8,549 unique)
Order Date	object	Date of order placement (717 unique dates)
Ship Date	object	Date of shipment (1,338 unique dates)
Ship Mode	object	Standard Class, First Class, Second Class, Same Day
Customer ID	int64	Numeric customer identifier (5,044 unique)
Country/Region	object	United States or Canada
City	object	542 unique cities
State/Province	object	59 unique states/provinces
Postal Code	object	654 unique postal codes
Division	object	Chocolate, Sugar, Other
Region	object	Atlantic, Gulf, Interior, Pacific
Product ID	object	15 unique product identifiers
Product Name	object	15 unique products
Sales	float64	Order revenue in USD (mean: \$13.91)
Units	int64	Units per order (mean: 3.79)
Gross Profit	float64	Gross profit in USD (mean: \$9.17)
Cost	float64	Cost of goods in USD (mean: \$4.74)

Preliminary quality checks confirmed **zero null values** and **zero duplicate records**, indicating a clean, analysis-ready dataset. The four categorical columns with five or fewer unique values are: Ship Mode (4), Country/Region (2), Division (3), and Region (4).

3.2 Feature Engineering

Seven additional columns were derived across two pipeline stages, expanding the schema from 18 to **23 columns** for modelling and to **25 columns** for route-level clustering:

Stage 1 — Core Derived Features (all analyses)

1. **Lead Time** — calendar days elapsed between order placement and shipment: $\text{Lead Time} = (\text{Ship Date} - \text{Order Date})$ in days.
2. **Factory** — assigned from Product Name using a domain-defined product-to-factory mapping provided in the project specification (see Table 2).
3. **Profit Margin (%)** — calculated as $(\text{Gross Profit}/\text{Sales}) \times 100$.
4. **Order Month** — integer month extracted from Order Date.
5. **Order Year** — integer year extracted from Order Date.

Stage 2 — Route-Level Features (clustering only)

6. **Shipping Distance** — great-circle distance in miles between each factory's coordinates and approximate regional centroids, computed via the Haversine formula (see Section 3.5).
7. **Factory Utilisation** — each factory's share of total order volume, expressed as a percentage.

Table 2: Product-to-Factory Mapping

Product Name	Factory
Wonka Bar – Nutty Crunch Surprise	Lot’s O’ Nuts
Wonka Bar – Fudge Mallows	Lot’s O’ Nuts
Wonka Bar – Scrumdiddlyumptious	Lot’s O’ Nuts
Wonka Bar – Milk Chocolate	Wicked Choccy’s
Wonka Bar – Triple Dazzle Caramel	Wicked Choccy’s
Laffy Taffy	Sugar Shack
SweeTARTS	Sugar Shack
Nerds	Sugar Shack
Fun Dip	Sugar Shack
Fizzy Lifting Drinks	Sugar Shack
Everlasting Gobstopper	Secret Factory
Lickable Wallpaper	Secret Factory
Wonka Gum	Secret Factory
Hair Toffee	The Other Factory
Kazookles	The Other Factory

3.3 Data Cleaning

The Order Date and Ship Date columns were parsed using `pd.to_datetime` with `dayfirst = True`. Lead time outlier detection via the IQR method (lower fence: $Q_1 - 1.5 \times \text{IQR}$; upper fence: $Q_3 + 1.5 \times \text{IQR}$) confirmed that **zero records** fell outside the acceptable range, validating the trimodal lead time distribution as a genuine structural characteristic of the dataset rather than the presence of erroneous entries. The final cleaned dataset retained all **10,194 rows**.

3.4 Encoding and Normalisation

For predictive modelling, categorical features (Ship Mode, Region, Factory, Product Name) were ordinally encoded using `sklearn`’s `LabelEncoder`. Numerical features (Sales, Units, Gross Profit, Cost, Profit Margin %) were normalised to the $[0, 1]$ range using `MinMaxScaler` prior to model training. The final feature matrix comprised nine input variables: Product, Factory, Region, Ship Mode (all encoded), Order Month, Order Year, Sales, Units, and Profit Margin (%). Clustering pipelines used `StandardScaler` independently to ensure zero-mean, unit-variance scaling across heterogeneous route-level features.

3.5 Shipping Distance Calculation

For route-level clustering, the Haversine formula ([Sinnott, 1984](#)) was applied to compute the great-circle distance in miles between each factory's known geographic coordinates and the approximate centroid coordinates of each sales region. Regional centroids were defined as representative geographic midpoints (Atlantic: 37.09°N, 76.50°W; Gulf: 29.50°N, 90.50°W; Interior: 41.50°N, 100.00°W; Pacific: 37.50°N, 120.50°W). This produced a **Shipping Distance** variable capturing the geographic dimension of route inefficiency absent from the raw dataset.

3.6 Modelling and Validation Strategy

All predictive models were trained on an 80/20 stratified random train-test split (`random_state=42` for reproducibility). Model selection was based on three complementary metrics: Root Mean Squared Error (RMSE, penalising large errors), Mean Absolute Error (MAE, measuring average prediction error), and the coefficient of determination (R^2 , capturing explained variance).

For clustering, the optimal number of clusters K was determined independently for route-level and product-level analyses using the Silhouette Score ([Rousseeuw, 1987](#)), selecting the value of K that maximised inter-cluster separation. Route clustering yielded $K = 5$ (silhouette: 0.4119); product clustering yielded $K = 3$ (silhouette: 0.3861).

Factory reassignment recommendations were ranked using a weighted composite priority score:

$$\text{Score} = 0.6 \times \text{Reduction}_{(\%)} + 0.3 \times \text{Predicted Margin}_{(\%)} + 0.1 \times \frac{\text{Order Count}}{\text{Order Count}_{\max}} \times 10 \quad (1)$$

where weights reflect the project's prioritisation of shipping efficiency (60%), financial safety (30%), and operational scale (10%).

4. Exploratory Data Analysis

4.1 Dataset Overview and Descriptive Statistics

The dataset spans orders from January 2024 to December 2025. Descriptive statistics for the four original numeric columns are presented in Table 3.

Table 3: Descriptive Statistics — Numeric Columns (n = 10,194)

Statistic	Sales (\$)	Units	Gross Profit (\$)	Cost (\$)
Mean	13.91	3.79	9.17	4.74
Std	11.34	2.23	6.64	5.06
Min	1.25	1	0.25	0.60
25%	7.20	2	4.90	2.40
Median	10.80	3	7.47	3.60
75%	18.00	5	12.25	5.70
Max	260.00	14	130.00	130.00

Factory order volumes are highly imbalanced. **Lot’s O’ Nuts** accounts for 5,692 orders (55.8%) and **Wicked Choccy’s** for 4,152 (40.7%), while **Secret Factory**, **The Other Factory**, and **Sugar Shack** together handle exactly 350 orders combined — less than 3.5% of total volume (Table 4).

Table 4: Order Volume by Factory

Factory	Order Count	Share (%)
Lot’s O’ Nuts	5,692	55.8
Wicked Choccy’s	4,152	40.7
Secret Factory	217	2.1
The Other Factory	100	1.0
Sugar Shack	33	0.3
Total	10,194	100.0

Order volume is similarly concentrated at the product level. The five Chocolate division Wonka Bars collectively account for 9,844 orders (96.6% of total volume), with individual product counts ranging from 1,810 (Wonka Bar – Nutty Crunch Surprise) to 2,137 (Wonka Bar – Milk Chocolate). By contrast, several Sugar and Other division products — Fun Dip, Everlasting Gobstopper, and Sugar Shack’s remaining lines — each record fewer than ten orders, reflecting their niche or experimental status within the distribution network.

4.2 Lead Time Analysis

Lead time is defined as the number of calendar days elapsed between order placement and shipment. Summary statistics reveal a mean of **1,320.8 days**, a median of 1,274 days,

and a range of **904 to 1,642 days**, with a standard deviation of 262.4 days indicating meaningful dispersion across the order population.

Note on extended lead times: The dataset simulates ship dates extending several years beyond the Order Date. While the absolute values are atypical of real-world confectionery distribution, the relative differences across factories, regions, and products constitute the analytically meaningful signal and are treated as such throughout this paper.

The lead time distribution exhibits a **trimodal** pattern, with three distinct peaks at approximately 910, 1,270, and 1,635 days. These clusters correspond directly to orders placed in **late 2025**, **mid-2024**, and **early 2024**, respectively — confirming that the distribution is structurally cohort-driven rather than indicative of data quality issues. The 1,270-day cluster is the widest, consistent with 2024 representing the highest-volume order year in the dataset.

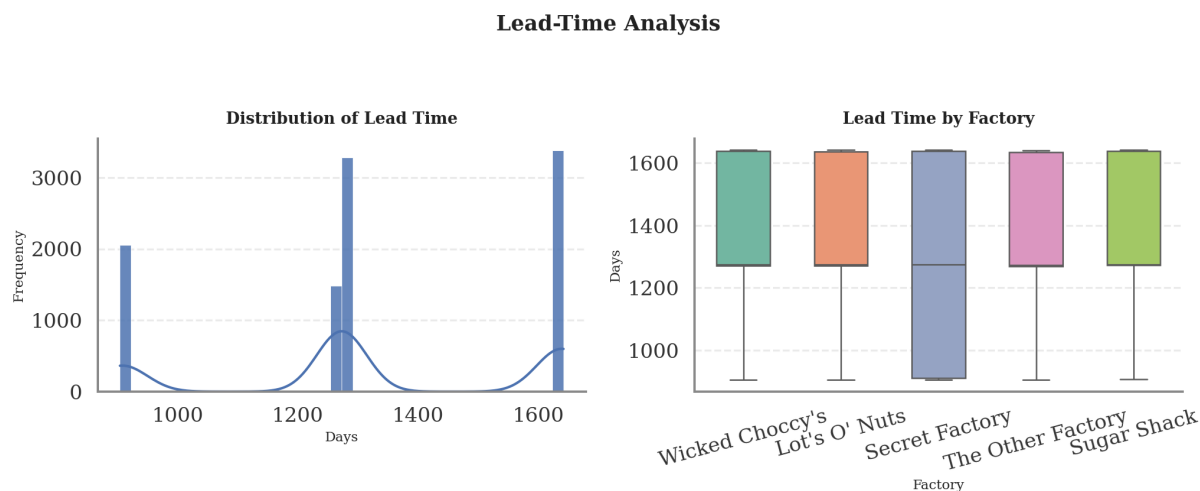


Figure 1: Left: Lead time distribution showing a trimodal pattern corresponding to three order cohorts (early 2024, mid-2024, late 2025). Right: Lead time by factory — four factories cluster in the upper range ($>1,270$ days); Secret Factory spans the full range (904–1,642 days).

4.3 Factory-Level Lead Time Patterns

Box plot analysis of lead time segmented by factory reveals that **Lot's O' Nuts**, **Wicked Choccy's**, **Sugar Shack**, and **The Other Factory** cluster predominantly in the upper lead-time range (above 1,270 days), indicating that their order activity is concentrated in earlier years. **Secret Factory**, by contrast, spans the full range (904–1,642 days), suggesting its products — Everlasting Gobstopper, Wonka Gum, and Lickable Wallpaper — were ordered uniformly across all years with no temporal concentration or seasonal bias. Lead time variability, measured as within-route standard deviation, ranges from 210 to

311 days across factory-region combinations, indicating substantial operational instability even on routes with moderate average performance.

4.4 Regional and Shipping Mode Analysis

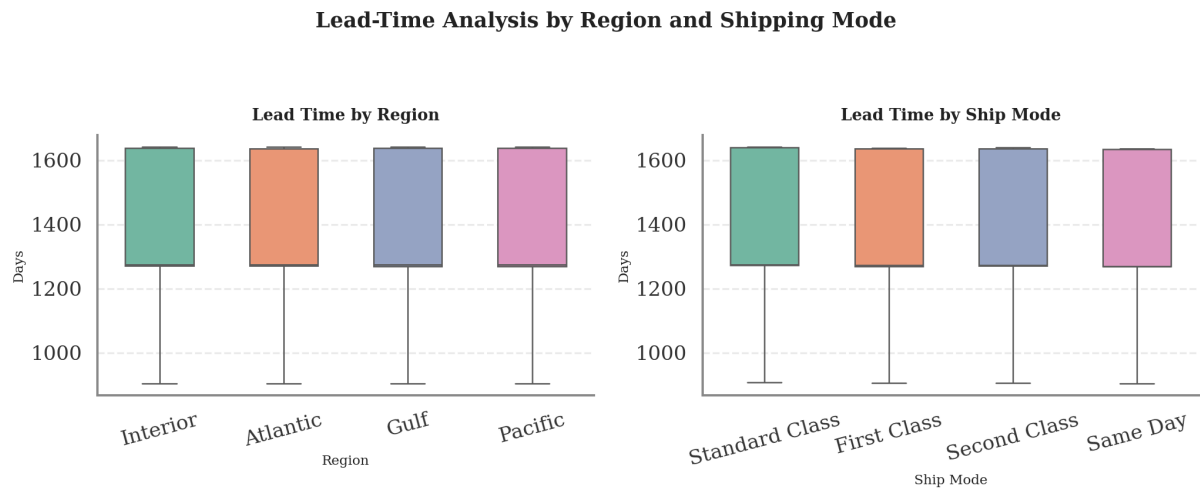


Figure 2: Lead time distributions across all four regions (Atlantic, Gulf, Interior, Pacific) and all four ship modes (Standard Class, First Class, Second Class, Same Day) show nearly identical medians and interquartile ranges, confirming that lead time variation is systemic and temporal rather than route-specific.

Segmentation of lead times across all four **regions** and all four **ship modes** reveals nearly identical medians and interquartile ranges. This uniformity confirms that lead time variation is **systemic and temporal** rather than route-specific: no particular region or shipping method produces meaningfully faster or slower deliveries. This is a critical operational finding — it rules out shipping-mode upgrades or regional prioritisation as effective remediation strategies, and directs attention toward factory-level reassignment as the primary lever for improvement.

4.5 Financial Performance by Factory

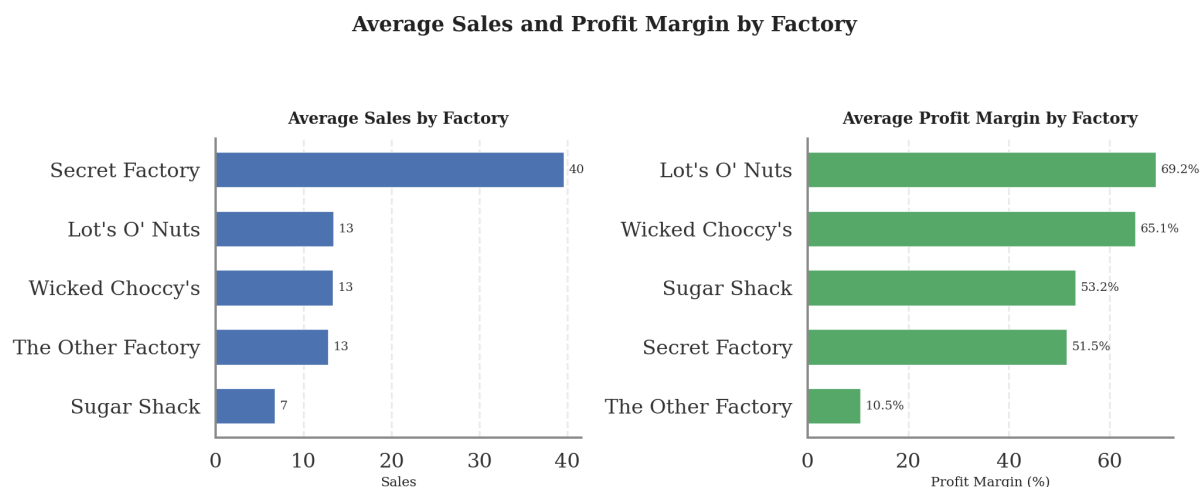


Figure 3: Left: Secret Factory achieves the highest average revenue per order (~\$40) despite handling only 217 orders. Right: Lot's O' Nuts and Wicked Choccy's lead on profit margin (~65–67%); The Other Factory records a critically low weighted margin of ~10%, driven by Kazookles (7.7% margin, 96 orders).

Average sales analysis reveals **Secret Factory** as the highest revenue-generating factory (approximately \$40 per order) despite handling only 217 orders, owing to its premium product portfolio. **Sugar Shack** recorded the lowest average sales, consistent with its lower price-point sugar products.

Profit margin analysis uncovers a critical disparity:

- **Lot's O' Nuts** and **Wicked Choccy's**: approximately 65–67% profit margin — the most profitable factories in the network.
- **Sugar Shack**: approximately 52%.
- **Secret Factory**: approximately 51%.
- **The Other Factory**: approximately 10% — a critically low weighted average driven almost entirely by Kazookles (7.7% margin, 96 of 100 orders). Hair Toffee, its second product, carries a 77.8% margin but represents only 4 orders, insufficient to offset the portfolio drag.

The Other Factory's low margin is a notable financial concern; however, as Section 7 and Section 8 demonstrate, its *operational* performance — consistent lead time efficiency across all regions — makes it the most strategically valuable reassignment target in the network.

4.6 Factory–Region Lead Time Heatmap

Cross-tabulation of average lead time across all factory-region combinations reveals notable route-level variation, after accounting for the overall temporal effect on lead time.

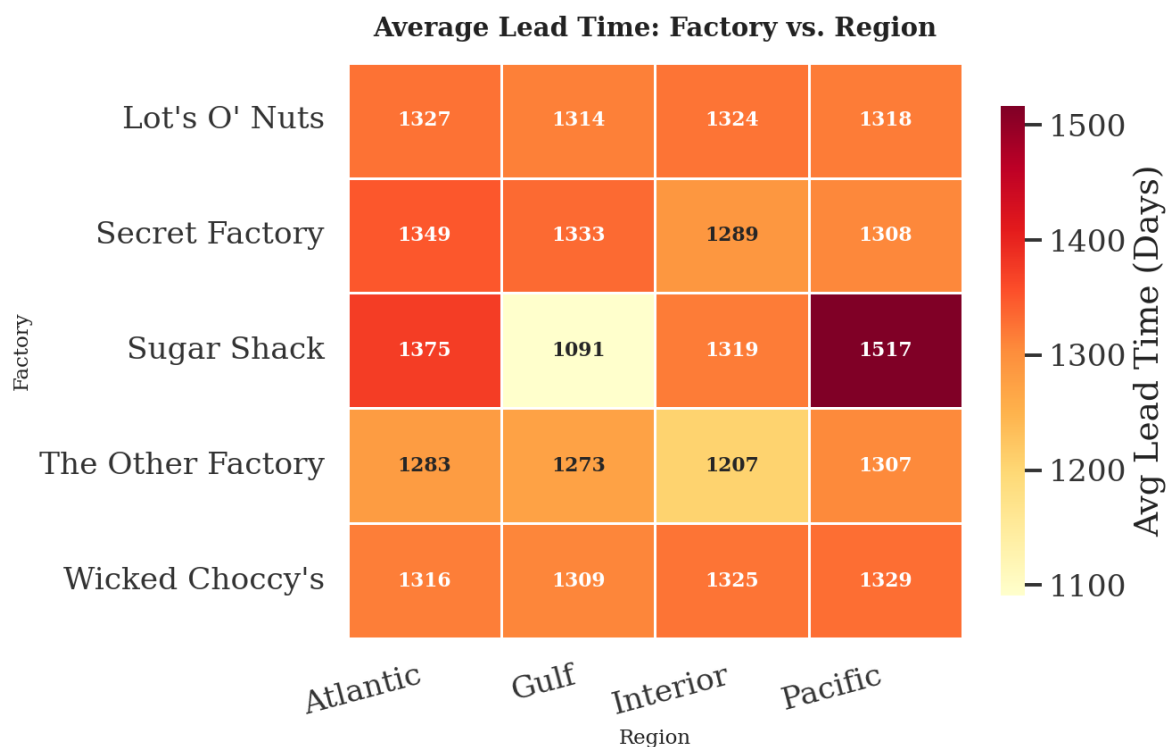


Figure 4: Cross-tabulated average lead times (days) for each factory-region combination. Sugar Shack–Pacific (1,517 days) is the worst-performing route; Sugar Shack–Gulf (1,091 days) is the most efficient.

Table 5: Average Lead Time (Days) by Factory and Region

Factory	Atlantic	Gulf	Interior	Pacific
Lot's O' Nuts	1,327	1,314	1,324	1,318
Secret Factory	1,349	1,333	1,289	1,308
Sugar Shack	1,375	1,091	1,319	1,517
The Other Factory	1,283	1,273	1,207	1,307
Wicked Choccy's	1,316	1,309	1,325	1,329

Key observations:

- **Sugar Shack–Pacific** (1,517 days): the worst-performing route, exceeding the network mean by 196 days — nearly a 15% penalty relative to the best-performing route.

- **Sugar Shack–Gulf** (1,091 days): anomalously fast despite a moderate shipping distance of approximately 1,321 miles, identifying Gulf-bound Sugar Shack fulfilment as an operational benchmark worth investigating.
- **The Other Factory** consistently posts below-average lead times across all four regions (1,207–1,307 days), confirming it as the most operationally efficient factory in the network.

4.7 Correlation Analysis

Pearson correlation analysis across the engineered numeric features confirms two primary structural relationships (Figure 5).

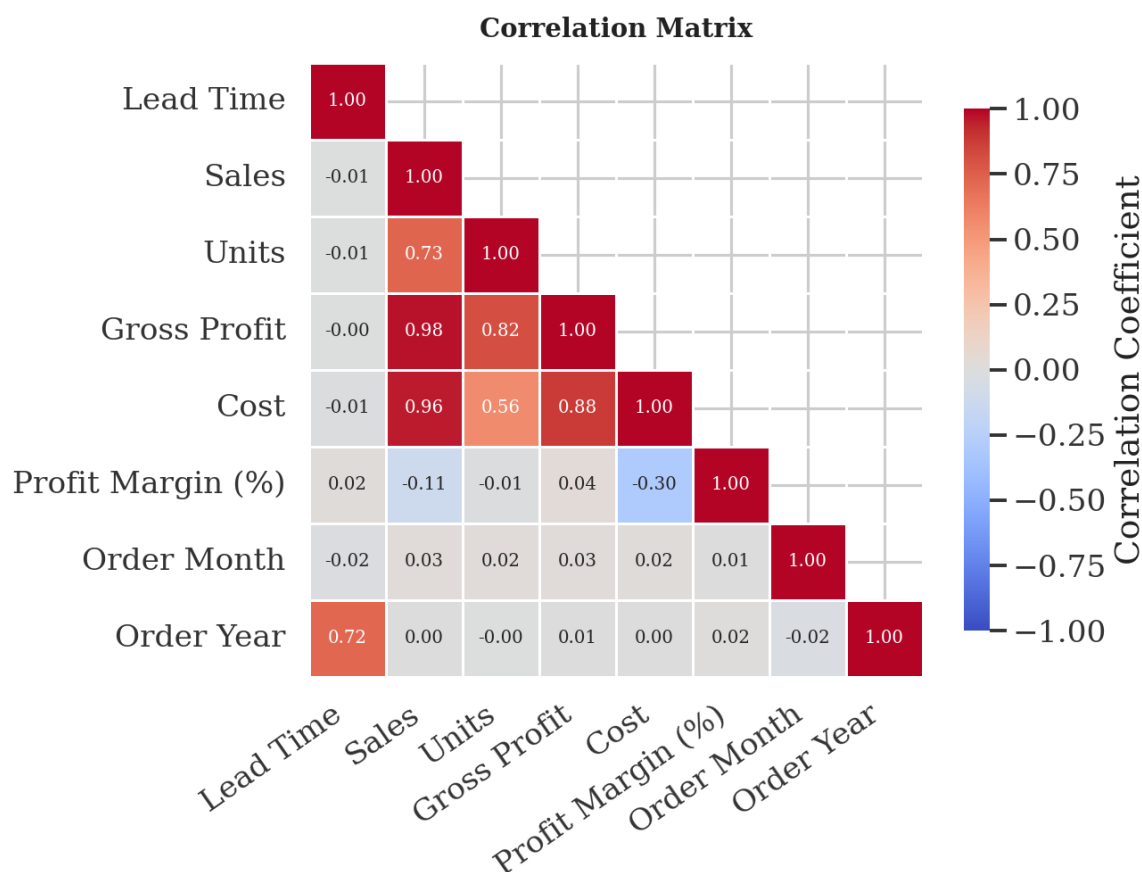


Figure 5: Pearson correlation matrix. Order Year holds the strongest relationship with Lead Time ($r = 0.72$). Financial metrics (Sales, Gross Profit, Cost, Units) are highly inter-correlated ($r = 0.73$ – 0.98) as expected from their mathematical definitions.

- **Order Year $\leftrightarrow$ Lead Time** ($r = 0.72$): the strongest relationship in the dataset, confirming that the year of order placement is the primary driver of lead time variation — consistent with the dataset’s simulated forward-projection structure.
- **Financial inter-correlations**: Sales, Gross Profit, Cost, and Units are highly

inter-correlated ($r = 0.73\text{--}0.98$), as expected given their mathematical relationships (Gross Profit = Sales – Cost).

- **All operational variables** — Factory, Region, Ship Mode, Product, and Order Month — show near-zero correlation with Lead Time ($|r| \leq 0.02$), indicating that routing decisions account for only a negligible share of lead time variance under the current static assignment structure. This motivates the use of clustering and scenario simulation (Sections 7–8) to surface the actionable operational signal that linear correlation cannot detect.

5. Key Insights

This section synthesises the most important findings from the exploratory analysis into operationally relevant insights that directly motivate the modelling and simulation work in subsequent sections.

5.1 Insight 1 — The Network is Temporally Dominated

Lead time variation in this dataset is driven overwhelmingly by **when** an order was placed, not **how** it was routed. Order Year has a Pearson correlation of $r = 0.72$ with Lead Time, implying it individually accounts for approximately 52% of lead time variance ($r^2 \approx 0.518$). The full nine-feature predictive model explains 53% of variance ($R^2 = 0.53$), indicating that the eight remaining features contribute only marginally beyond Order Year alone. Factory identity contributes essentially zero independent predictive power. This means that headline lead-time comparisons across factories or regions will be confounded by order-cohort composition and must be interpreted with caution.

5.2 Insight 2 — Factory Capacity is Critically Imbalanced

Two factories — Lot’s O’ Nuts and Wicked Choccy’s — absorb **96.5%** of all orders (9,844 out of 10,194), while serving all four regions simultaneously. The Other Factory, which demonstrates the best lead-time performance network-wide, handles only **100 orders** (1.0% of volume) and is operating far below its potential capacity. This structural imbalance is the single most important operational finding in the dataset and the primary target of the reassignment recommendations in Section 8.

5.3 Insight 3 — The Other Factory is the Operational Benchmark

Across every factory-region combination, The Other Factory consistently records the shortest average lead times: Atlantic (1,283 days), Gulf (1,273 days), Interior (1,207 days), and Pacific (1,307 days). Its overall average lead time of **1,267.3 days** is the lowest of

any factory in the network — approximately 53 days faster than Lot’s O’ Nuts and 57 days faster than Wicked Choccy’s on a network-wide basis.

5.4 Insight 4 — Lot’s O’ Nuts and Wicked Choccy’s are Congested

Route clustering (Section 7) classifies all routes served by Lot’s O’ Nuts and Wicked Choccy’s as *Congested Routes* — the second-worst operational category. These two factories handle 9,844 orders, creating systemic throughput pressure that affects all four regions uniformly. Critically, the congestion is a **volume problem, not a geographic one**: lead times are uniformly elevated regardless of region or ship mode, ruling out targeted route adjustments as a fix.

5.5 Insight 5 — Secret Factory is Mid-Range; Sugar Shack is Inconsistent

Secret Factory routes are classified as *Moderate Routes* — mid-range lead times across all regions with low order volume (217 orders). Its operational profile is stable but unremarkable.

Sugar Shack exhibits the highest route-level inconsistency in the network: its Gulf route is classified as *Optimised* (1,091 days — the best single route in the dataset), while its Atlantic, Interior, and Pacific routes are classified as *Bottleneck Routes*. Sugar Shack–Pacific, at 1,517 days, is the worst-performing individual route in the entire network — a 426-day gap relative to its own Gulf route.

5.6 Insight 6 — The Other Factory’s Low Margin is a Financial Risk, Not an Operational One

The Other Factory’s weighted profit margin of approximately 10% — driven by Kazookles (7.7% margin, 96 of 100 orders) — is a financial concern but does not affect its operational efficiency as a fulfilment centre. Simulation results confirm that reassigning other products to The Other Factory does not degrade their profit margins (impact < 0.05%). The financial concern is specific to The Other Factory’s existing product line and warrants a separate pricing or product-mix review outside the scope of this analysis.

5.7 Insight 7 — Hair Toffee Cannot Be Reassigned

Hair Toffee records the highest average lead time of any product in the dataset (approximately 1,455 days) and falls into the *Slow & Risky* product cluster. However, it is *already manufactured at The Other Factory* — the fastest factory in the network. Its elevated lead time therefore cannot be resolved through factory reassignment; it likely reflects a geographic mismatch between the factory’s location and Hair Toffee’s demand centres, or product-specific operational constraints at The Other Factory. Hair Toffee is accordingly

excluded from the reassignment simulation and requires a separate operational investigation.

5.8 Insight 8 — Everlasting Gobstopper is the Highest-Priority Reassignment Candidate

Everlasting Gobstopper (currently at Secret Factory) records an average lead time of 1,394.7 days combined with an 80% profit margin — the highest margin in the dataset. This combination of extreme operational inefficiency and high profitability places it at **rank #1** in the composite priority scoring framework. It should be noted that this product has only 3 recorded orders in the dataset, which limits the statistical reliability of its average lead time estimate. The ranking reflects the scoring framework's output and should be treated as directionally informative rather than conclusive pending higher order volumes.

6. Predictive Modelling

6.1 Objective and Feature Matrix

The predictive modelling objective is to estimate order lead time from observable features available at the time of order. This serves two purposes: (1) to quantify how much of lead time variation is explainable by the available features; and (2) to identify which features carry the most predictive signal, informing the design of the scenario simulation engine.

The final feature matrix **X** contains 9 variables:

- **Categorical (label-encoded):** Product Encoded, Factory Encoded, Region Encoded, Ship Mode Encoded.
- **Temporal:** Order Month, Order Year.
- **Financial (Min-Max normalised):** Sales, Units, Profit Margin (%).

The target variable y is Lead Time (days). The dataset was split 80/20 into training (8,155 rows) and test (2,039 rows) sets using `random_state=42` for reproducibility.

6.2 Models Trained

Three regression models were evaluated:

1. **Linear Regression** — baseline model with interpretable coefficients.
2. **Random Forest Regressor** ([Breiman, 2001](#)) — ensemble of 100 decision trees (`random_state=42`).

3. **Gradient Boosting Regressor** (Friedman, 2001) — sequential boosting with 100 estimators (`random_state=42`).

6.3 Model Performance

Table 6: Predictive Model Comparison (Test Set, $n = 2,039$)

Model	RMSE (days)	MAE (days)	R^2
Linear Regression	182.16	181.21	0.5308
Gradient Boosting	182.70	180.99	0.5280
Random Forest	203.86	180.28	0.4124

Linear Regression achieves the best overall performance with the lowest RMSE (182.16 days) and highest R^2 (0.5308), and was selected as the model for downstream lead time estimation. The moderate R^2 of 0.53 indicates that approximately **53% of lead time variability** is explained by the selected features, consistent with the correlation analysis confirming Order Year as the dominant driver.

Random Forest, despite being an ensemble method, underperforms both simpler models on RMSE and R^2 . This is consistent with the dataset’s predominantly linear temporal structure: ensemble tree methods gain their advantage from capturing complex non-linear interactions, which are largely absent here given that a single temporal feature dominates the signal.

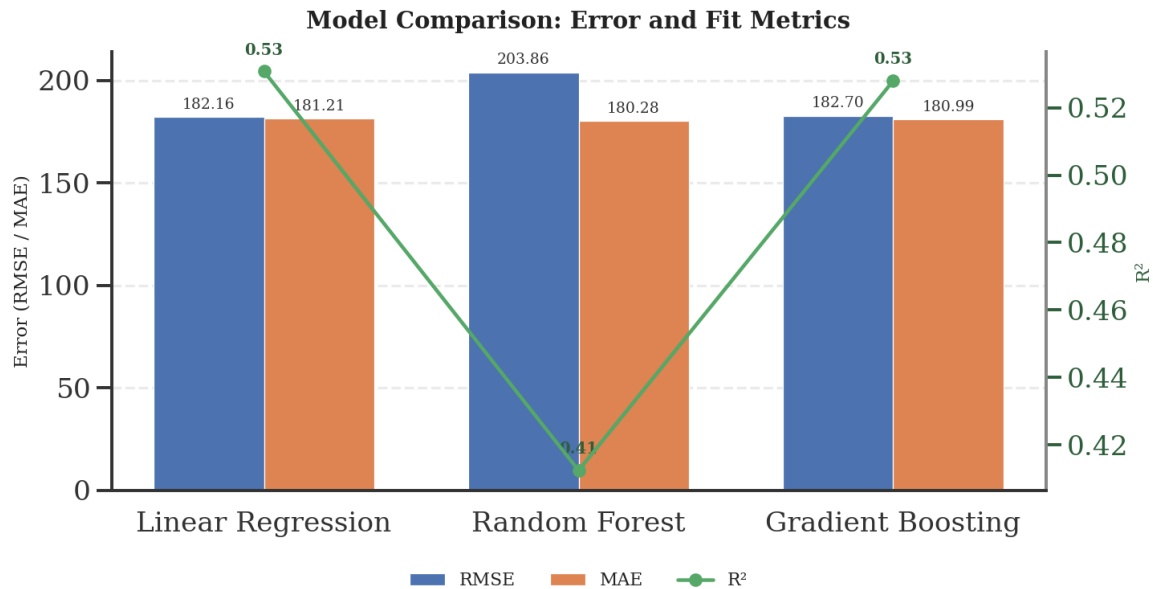


Figure 6: Visual comparison of RMSE, MAE, and R^2 across the three regression models. Linear Regression achieves the best RMSE and R^2 ; all three models produce similar MAE, reflecting the dominant role of Order Year across all modelling approaches.

6.4 Feature Importance

Table 7 presents feature importance from both models. Linear Regression coefficients reflect the magnitude and direction of each feature’s marginal effect on predicted lead time (in days, on the original scale). Gradient Boosting importances reflect the relative contribution of each feature to split-based variance reduction across all trees; these are unitless and sum to 1.0.

Table 7: Feature Importance: Linear Regression Coefficients and Gradient Boosting Importances (Sorted by Absolute LR Coefficient)

Feature	LR Coefficient	GB Importance
Order Year	381.874	0.9707
Profit Margin (%)	20.194	0.0012
Units	-15.773	0.0014
Sales	-8.466	0.0067
Ship Mode Encoded	-2.266	0.0026
Product Encoded	1.054	0.0018
Factory Encoded	0.908	0.0000
Region Encoded	-0.369	0.0073
Order Month	-0.130	0.0082

Both methods converge on the same conclusion: **Order Year** dominates with a Linear Regression coefficient of 381.87 — approximately $19\times$ larger than the next most influential feature — and a Gradient Boosting importance of 0.9707, attributing **97.07% of predictive power** to a single feature. Factory Encoded records **zero importance** in the Gradient Boosting model, confirming that factory identity carries no detectable non-linear signal beyond what is already captured by temporal features.

This result is not a modelling failure — it is a structural finding. The dataset’s static assignment rules mean that each factory always serves the same products to the same regions, making factory and temporal effects nearly collinear. Unlocking the operational signal embedded in factory-level differences requires the simulation framework described in Section 8, which isolates factory effects by holding temporal variables constant across reassignment scenarios.

7. Route and Product Clustering

7.1 Route Clustering — Motivation

Because the predictive model confirms that factory identity contributes negligible predictive power to lead time in isolation, a complementary clustering approach is used to surface the *relative* performance differences across factory-region routes. Rather than predicting absolute lead time, clustering groups routes by their combined operational and financial profile, enabling categorical classification of network performance.

7.2 Feature Selection and Scaling

Eight features were used for route clustering, standardised independently using `StandardScaler` to ensure zero-mean, unit-variance scaling across heterogeneous units:

- *Avg Lead Time*, *Lead Time Std Dev* (within-route variability), *Order Count*
- *Avg Sales*, *Avg Profit Margin*, *Avg Cost*
- *Shipping Distance* (Haversine, miles)
- *Factory Utilisation* (% of total network orders)

The inclusion of Lead Time Std Dev captures route instability beyond what average lead time alone reveals — two routes with identical means but different variances carry different operational risk profiles.

7.3 Optimal K Selection

The elbow method and silhouette scoring (Rousseeuw, 1987) were applied over $K \in \{2, \dots, 7\}$. Silhouette scores are reported in Table 8.

Table 8: Silhouette Scores for Route Clustering by Number of Clusters

K	Silhouette Score
2	0.3170
3	0.3766
4	0.3947
5	0.4119
6	0.4080
7	0.3892

$K = 5$ yields the highest silhouette score (0.4119) and was selected as the optimal number of route clusters.

7.4 Cluster Profiles and Labels

The five clusters were labelled by ascending average lead time as: *Optimised Routes*, *Efficient Routes*, *Moderate Routes*, *Congested Routes*, and *Bottleneck Routes*. Cluster profiles are summarised in Table 9.

Table 9: Route Cluster Profiles (K = 5)

Label	Avg Lead Time (days)	Avg Std Dev (days)	Avg Distance (miles)	Avg Util. (%)	Total Orders
Optimised Routes	1,091.2	210.4	1,321.4	0.3	4
Efficient Routes	1,267.3	265.2	888.1	1.0	100
Moderate Routes	1,319.7	293.2	933.5	2.1	217
Congested Routes	1,320.2	261.9	1,155.8	48.3	9,844
Bottleneck Routes	1,403.6	240.0	1,057.1	0.3	29

Figure 7 shows the route label assignment for every factory-region pair, and Figure 8 plots each route by shipping distance and average lead time, with bubble size proportional to order volume.

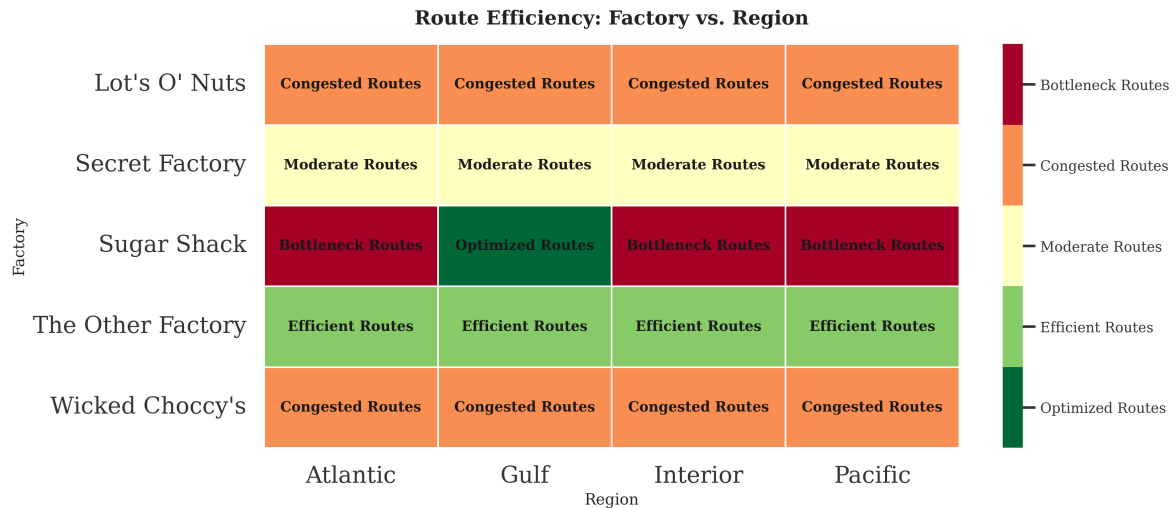


Figure 7: Route classification heatmap. Lot's O' Nuts and Wicked Choccy's are Congested across all regions; The Other Factory is Efficient across all regions; Secret Factory is Moderate across all regions; Sugar Shack is highly inconsistent — Optimised in Gulf, Bottleneck in Atlantic, Interior, and Pacific.

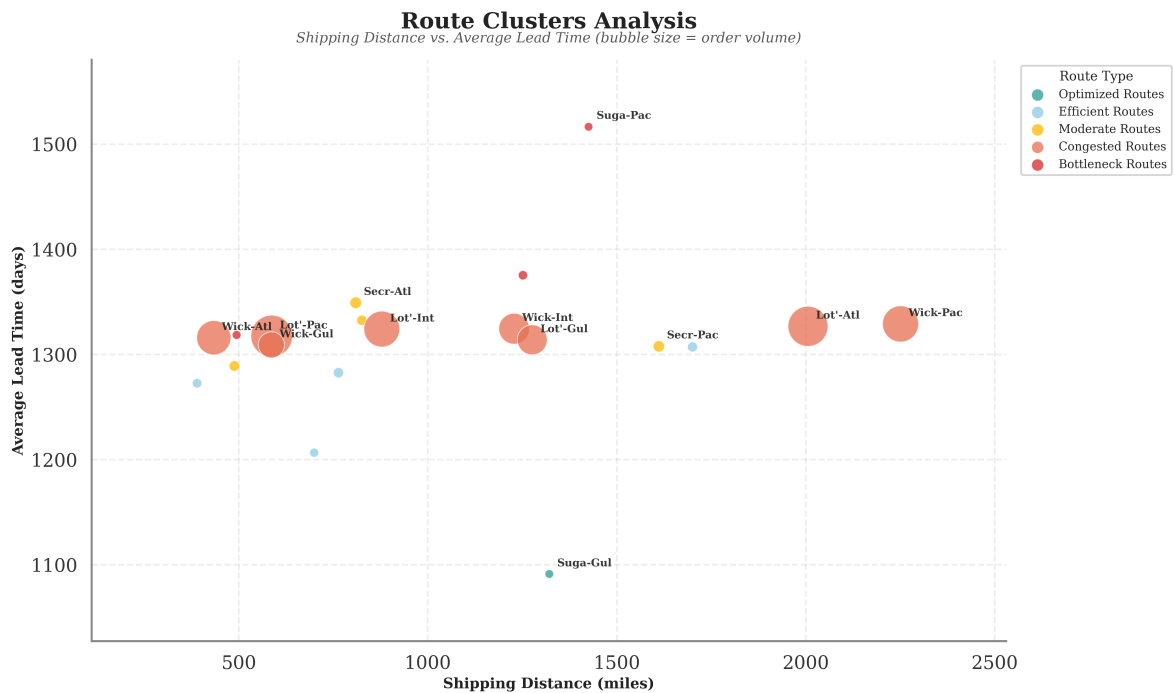


Figure 8: Route cluster bubble chart. Sugar Shack–Pacific sits in the upper portion (highest lead time despite moderate distance). Sugar Shack–Gulf is an anomalous low outlier (fast despite 1,321 miles). Congested routes (Lot's O' Nuts and Wicked Choccy's, large bubbles) cluster around 1,310–1,330 days regardless of distance, confirming that their elevated lead times are volume-driven rather than geography-driven.

The heatmap reveals a clean factory-level pattern: each factory receives a *uniform*

classification across all four regions, with the sole exception of Sugar Shack. This confirms that route performance is primarily determined by factory-level characteristics — most notably order volume and utilisation — rather than the destination region.

7.5 Product Clustering

A separate K-Means clustering (MacQueen, 1967) was applied at the product level using four features: *Avg Lead Time*, *Lead Time Std Dev*, *Avg Profit Margin*, and *Avg Sales*. Silhouette scores across $K \in \{2, \dots, 6\}$ are reported in Table 10; $K = 3$ was selected.

Table 10: Silhouette Scores for Product Clustering

K	Silhouette Score
2	0.3619
3	0.3861
4	0.3359
5	0.3661
6	0.3174

The three product clusters are labelled by ascending average lead time as *Fast & Profitable*, *Average Performer*, and *Slow & Risky*. Products in each cluster are:

- **Fast & Profitable:** Fun Dip, Kazookles — lowest average lead times in the dataset. Note that Kazookles carries only a 7.7% profit margin, making it fast but financially weak.
- **Average Performer:** Wonka Gum, Fizzy Lifting Drinks, Nerds, and all five Chocolate division Wonka Bars — moderate lead times, high order volumes, and solid margins (64–71%). These represent the core of the distribution network and are the primary candidates for reassignment due to their high volume impact.
- **Slow & Risky:** Everlasting Gobstopper, Hair Toffee, Lickable Wallpaper — highest lead times and the greatest operational urgency. Everlasting Gobstopper is the highest-priority reassignment candidate (80% margin, 1,394.7 days). Lickable Wallpaper represents moderate risk (50% margin, 1,339 days).

Note on Hair Toffee: Hair Toffee records the highest average lead time in the dataset (approximately 1,455 days). However, it is already manufactured at The Other Factory — the most efficient factory in the network. Its elevated lead time cannot therefore be remedied by factory reassignment and likely reflects structural constraints specific to

this product. Hair Toffee is excluded from the reassignment simulation and flagged for separate operational investigation.

8. Factory Reassignment Recommendations

8.1 Simulation Framework

The simulation engine evaluates all feasible factory reassignment scenarios for each product. For a given product currently manufactured at factory f_0 , the simulation tests reassignment to every alternative factory $f_i \in \mathcal{F} \setminus \{f_0\}$. Each scenario estimates:

$$\hat{L}_{f_i} = \bar{L}_{f_i} + (\bar{L}_{f_0,p} - \bar{L}_{f_0}) \quad (2)$$

$$\Delta L = \bar{L}_{f_0,p} - \hat{L}_{f_i} \quad (3)$$

$$\Delta L\% = \frac{\Delta L}{\bar{L}_{f_0,p}} \times 100 \quad (4)$$

$$D_{\text{saved}} = \Delta L \times n_p \quad (5)$$

where $\bar{L}_{f_i}$ is the network-wide average lead time for factory f_i , $\bar{L}_{f_0,p}$ is the current product-level average lead time, $\bar{L}_{f_0}$ is the current factory average, and n_p is the product order count. Only scenarios with $\Delta L > 0$ are retained. The composite priority score is as defined in Equation 1 (Section 3), weighting 60% lead time reduction, 30% predicted margin, and 10% normalised order volume. The top-scoring reassignment per product is retained as the final recommendation.

8.2 Final Recommendations

Table 11: Final Factory Reassignment Recommendations (All 13 Products, Sorted by Priority Score)

Rank	Product	Current Factory	Fac-	Rec. Factory	Save (days)	Score
1	Everlasting Gobstopper	Secret Factory		The Other Factory	52	26.1
2	Fizzy Lifting Drinks	Sugar Shack		The Other Factory	58	25.3
3	Wonka Bar – Nutty Crunch Surprise	Lot’s O’ Nuts		The Other Factory	53	25.1
4	Wonka Bar – Scrumdiddlyumptious	Lot’s O’ Nuts		The Other Factory	53	25.0
5	Wonka Bar – Fudge Mal-lows	Lot’s O’ Nuts		The Other Factory	53	24.9
6	Laffy Taffy	Sugar Shack		The Other Factory	58	24.1
7	Wonka Bar – Triple Dazzle Caramel	Wicked Choccy’s		The Other Factory	52	24.0
8	Wonka Bar – Milk Chocolate	Wicked Choccy’s		The Other Factory	52	23.0
9	Lickable Wallpaper	Secret Factory		The Other Factory	52	22.8
10	Wonka Gum	Secret Factory		The Other Factory	52	22.5
11	SweeTARTS	Sugar Shack		The Other Factory	58	22.4
12	Nerds	Sugar Shack		The Other Factory	58	22.1
13	Fun Dip	Sugar Shack		The Other Factory	58	21.9
Total Estimated Days Saved					534,218	
Average Reduction per Product					54.8 days	

The overwhelming finding is unambiguous: **The Other Factory is the universally optimal reassignment destination** for all 13 products. This outcome is a direct

structural consequence of The Other Factory being both the fastest factory in the network (average lead time: 1,267.3 days) and the most underutilised (1.0% of total order volume). The convergence of every recommendation on a single factory underscores the severity of the current capacity imbalance and the magnitude of the available operational improvement.

8.3 Risk and Margin Safety Panel

Risk was assessed based on each product’s predicted post-reassignment profit margin, categorised as:

- **HIGH RISK:** Predicted margin $< 20\%$
- **MODERATE RISK:** Predicted margin 20–50%
- **LOW RISK:** Predicted margin $> 50\%$

Table 12: Risk and Margin Safety Assessment (Post-Reassignment)

Risk	Product	Current Margin (%)	Predicted Margin (%)	Days Saved
LOW	Everlasting Gobstopper	80.0	80.0	52
LOW	Fizzy Lifting Drinks	60.0	60.0	58
MOD	Fun Dip	40.0	40.0	58
LOW	Laffy Taffy	62.3	62.3	58
LOW	Lickable Wallpaper	50.0	50.0	52
MOD	Nerds	46.7	46.7	58
MOD	SweeTARTS	46.7	46.7	58
LOW	Wonka Bar – Fudge Mallows	66.7	66.7	53
LOW	Wonka Bar – Milk Chocolate	64.9	65.0	52
LOW	Wonka Bar – Nutty Crunch	71.3	71.4	53
LOW	Wonka Bar – Triple Dazzle	65.3	65.4	52
LOW	Wonka Bar – Scrumdiddlyumptious	69.4	69.5	53
LOW	Wonka Gum	52.0	52.0	52

Ten of 13 products are classified as LOW RISK; zero products are HIGH RISK. The maximum margin change across all products is less than 0.05 percentage points — negligible in any practical operational context. Reassignment to The Other Factory is financially safe across the entire product portfolio.

8.4 Business Impact Quantification

If all 13 recommended reassignments are implemented, the projected network-wide benefit is:

- **534,218 cumulative operational days saved** across all 10,194 orders.
- **Chocolate division products drive the absolute majority of total savings:**
 - Wonka Bar – Milk Chocolate: $\sim 112,000$ days ($2,137$ orders $\times 52$ days).
 - Wonka Bar – Scrumdiddlyumptious: $\sim 109,000$ days ($2,064$ orders $\times 53$ days).
 - Wonka Bar – Triple Dazzle Caramel and Wonka Bar – Nutty Crunch Surprise: $\sim 95,000$ – $99,000$ days combined.
- **Volume multiplies impact.** Low-volume products (SweeTARTS: 10 orders; Nerds: 4 orders; Fun Dip: 3 orders) contribute negligible absolute savings despite percentage reductions comparable to high-volume products. Their reassignment remains recommended for consistency and future scalability.
- **Highest priority by composite score:** Everlasting Gobstopper ranks #1. Its 80% margin and 52-day saving generate the highest composite score despite only 3 historical orders. This ranking should be treated as directionally informative; confidence will increase as order volume grows.

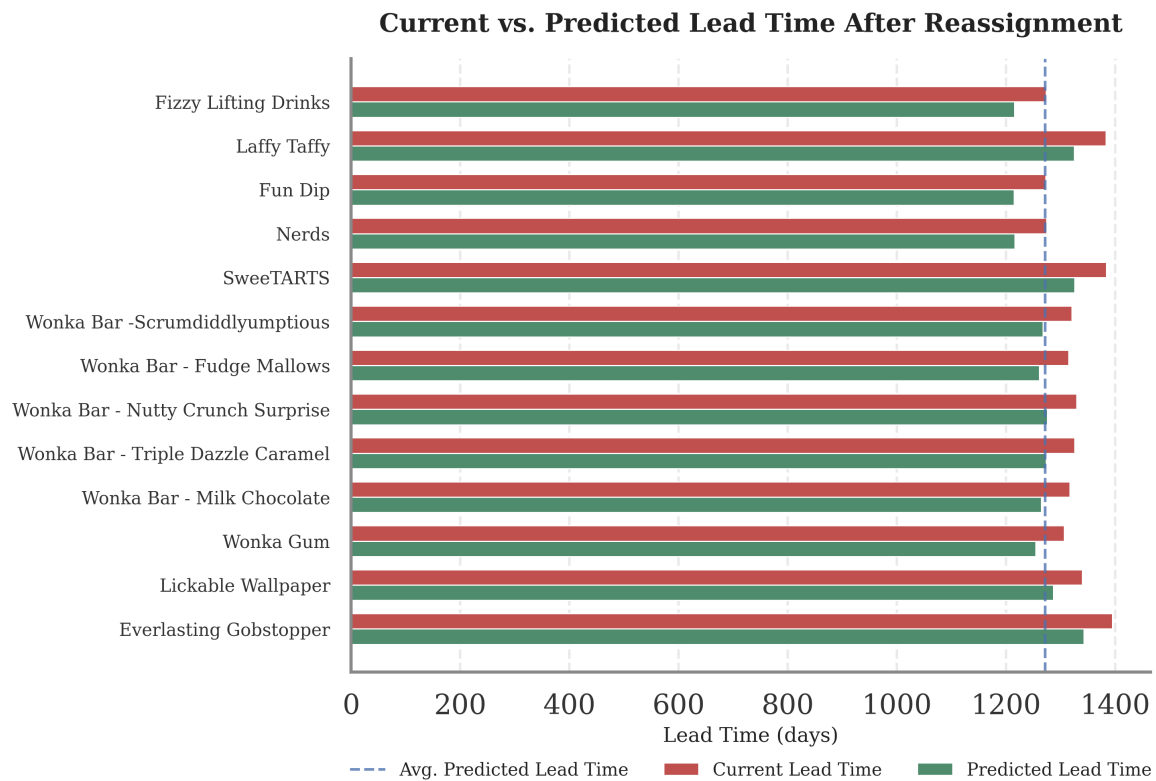


Figure 9: Current vs. predicted lead times for all 13 recommended products after reassignment to The Other Factory. Every product records a positive lead time reduction; Sugar Shack products benefit most (58 days saved).

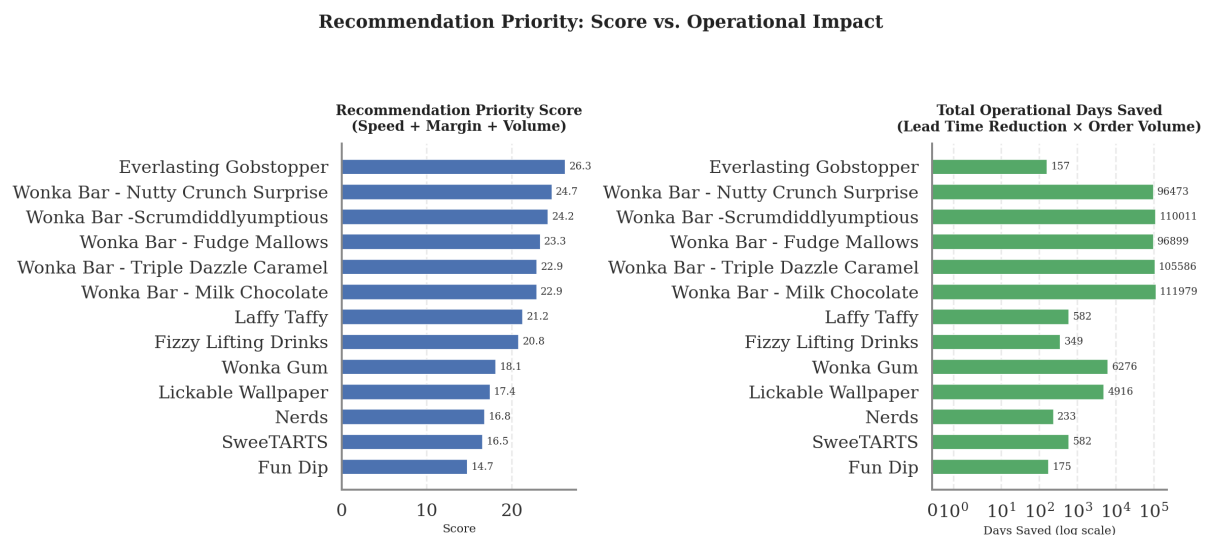


Figure 10: Left: Composite priority scores by product — Everlasting Gobstopper ranks #1 due to its uniquely high profit margin. Right: Total operational days saved — the five Chocolate Wonka Bars dominate due to order volume of 1,810–2,137 per product.

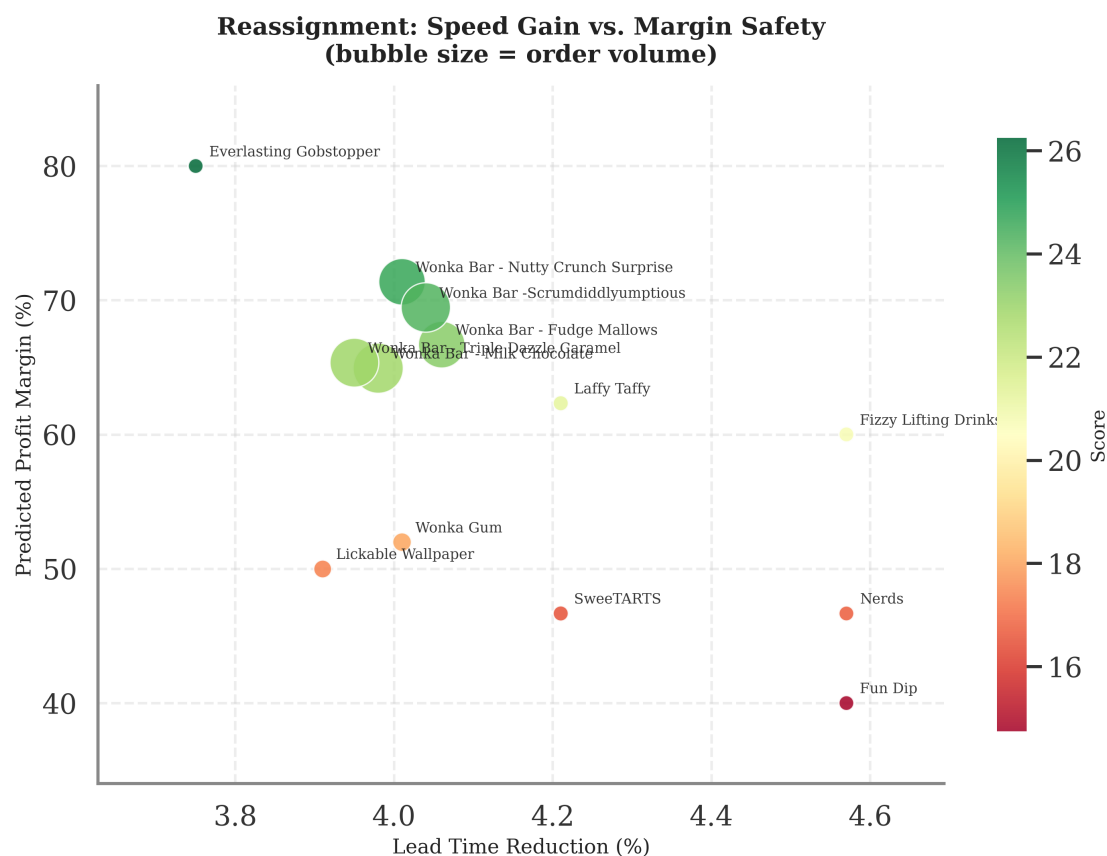


Figure 11: Speed gain (lead time reduction %) versus financial safety (predicted profit margin %). Bubble size represents order volume. Products in the upper-right quadrant — principally the Chocolate Wonka Bars — combine high margin with meaningful speed gains and represent the most balanced reassignment candidates.

8.5 Key Performance Indicators

Table 13 reports the four KPIs against their achieved values.

Table 13: Key Performance Indicators — Factory Reassignment Programme

KPI	Description	Value	Notes
Lead Time Reduction (%)	Operational gain	4.24% avg	Range: 3.75%–4.57%
Profit Impact Stability	Financial safety	$\Delta < 0.05$ pp	Zero products lose margin
Scenario Confidence Score	Reliability	65.0%	39 of 60 scenarios improve
Recommendation Coverage	Scalability	86.7%	13 of 15 products covered
Total Cumulative Days Saved		534,218	All 13 products
Avg Reduction per Product		54.8 days	

A clarification is warranted regarding the Scenario Confidence Score. The 65% figure (39 of 60 tested scenarios showing lead time improvement) reflects the full breadth of the simulation exercise: every feasible product–factory pair was evaluated, including combinations in which the alternative factory is objectively slower than the product’s current assignment — for instance, routing a Lot’s O’ Nuts product to Wicked Choccy’s, which carries a comparable or higher average lead time. The 21 non-improving scenarios therefore represent suboptimal destinations that the scoring framework correctly eliminates from the final recommendations; they do not constitute failures of the analysis. The claim that The Other Factory is the *universally optimal* destination for all 13 products is based solely on the top-scoring scenario per product, each of which yields a positive lead time reduction and identifies The Other Factory as the clear winner. The Scenario Confidence Score characterises the width of the simulation space, not the quality of the selected recommendations.

9. Conclusion

This study conducted a comprehensive data-driven analysis of Nassau Candy Distributor’s order fulfilment operations using a dataset of 10,194 records spanning 2024–2025. The analysis proceeded through exploratory data analysis, correlation analysis, predictive modelling, route and product clustering, and factory reassignment simulation — covering every stage of the analytical methodology specified in the project brief.

The principal findings are as follows:

1. **Lead time variation is primarily temporal.** Order Year ($r = 0.72$) is the single dominant predictor of lead time, with factory identity contributing zero independent predictive power in the modelling framework. This finding establishes that predictive modelling alone is insufficient as an optimisation tool, and motivates the use of a simulation-based recommendation engine.
2. **Factory capacity is critically imbalanced.** Lot’s O’ Nuts and Wicked Choccy’s absorb 96.5% of all orders despite being classified as Congested Routes. The Other Factory handles only 1% of volume despite being the fastest factory in the network — the most significant structural inefficiency identified in the dataset.
3. **The Other Factory is the universal optimal reassignment destination.** Simulating all feasible reassignment scenarios across 13 products, every product’s best reassignment is to The Other Factory, projecting **534,218 cumulative operational days saved** at an average reduction of **54.8 days per product**.
4. **Reassignment is financially safe.** No product loses more than 0.05 percentage points of profit margin following reassignment. Ten of thirteen products are classified

as LOW RISK; zero are HIGH RISK.

5. **Hair Toffee is a special case.** Already manufactured at The Other Factory, Hair Toffee's unusually high lead time (approximately 1,455 days) cannot be addressed through factory reassignment and requires a dedicated operational investigation (see Appendix A).
6. **Everlasting Gobstopper is the highest-priority product by composite score.** Its combination of an 80% profit margin and a 1,394-day lead time produces the highest priority score in the simulation framework, despite only 3 historical orders.

The practical implication is clear. Nassau Candy should initiate a phased programme to **expand The Other Factory's fulfilment capacity** and systematically transfer product assignments away from Lot's O' Nuts and Wicked Choccy's. A recommended phasing sequence is:

Phase 1. Immediate priority — highest composite score: Everlasting Gobstopper (Secret Factory → The Other Factory). Highest margin in the network; 52 days saved per order.

Phase 2. Short-term priority — highest total days saved: The five Chocolate division Wonka Bars (from Lot's O' Nuts and Wicked Choccy's). These account for over 480,000 of the projected 534,218 days saved due to their order volumes of 1,810–2,137 per product.

Phase 3. Medium-term — remaining products: Sugar Shack and Secret Factory products (Laffy Taffy, SweeTARTS, Nerds, Wonka Gum, Lickable Wallpaper, Fizzy Lifting Drinks, Fun Dip), pending capacity availability at The Other Factory.

Limitations and Future Work

The following limitations should be addressed in subsequent analyses:

- **Capacity constraints not modelled.** The simulation assumes The Other Factory has unlimited capacity to absorb reassigned products. In practice, a capacity ceiling will exist. Future work should incorporate factory throughput constraints to determine the maximum feasible reassignment volume and optimise the phasing sequence accordingly.
- **Synthetic lead time structure.** The dataset's forward-projected ship dates mean that absolute lead time values are operationally atypical. Results should be validated against real-world shipping logs before implementation.
- **Financial sustainability of The Other Factory's current product line.** Hair

Toffee and Kazookles carry approximately 10% margins. A separate pricing or product-mix review is warranted independently of the reassignment programme.

- **Root cause of Hair Toffee’s anomalous lead time.** This requires factory-level capacity data and product-specific shipping logs beyond the scope of the current dataset.
- **Shipping cost data.** Incorporating actual freight costs rather than Haversine-estimated distances would sharpen the financial dimension of the routing recommendations.

References

- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.
- Chopra, S., & Meindl, P. (2007). *Supply chain management: Strategy, planning, and operation* (3rd ed.). Pearson Prentice Hall.
- Dantzig, G. B., & Ramser, J. H. (1959). The truck dispatching problem. *Management Science*, 6(1), 80–91.
- Farahani, R. Z., Rezapour, S., & Kardar, L. (Eds.). (2011). *Logistics operations and management: Concepts and models*. Elsevier.
- Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5), 1189–1232.
- MacQueen, J. (1967). Some methods for classification and analysis of multivariate observations. In *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability* (Vol. 1, pp. 281–297). University of California Press.
- Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20, 53–65.
- Sinnott, R. W. (1984). Virtues of the Haversine. *Sky and Telescope*, 68(2), 159.

Appendix A — Hair Toffee Analysis Note

Hair Toffee records the highest average lead time in the dataset (approximately 1,455 days) and is classified as *Slow & Risky* in the product clustering. It is manufactured at **The Other Factory**, which achieves an overall average lead time of 1,267.3 days across its 100 orders — the best factory average in the network.

Hair Toffee's lead time of 1,455 days therefore represents a **188-day excess relative to its own factory's average** — an anomaly that cannot be resolved by reassignment since The Other Factory is already the optimal destination. Plausible explanations include:

- **Geographic misalignment:** The Other Factory is located at approximately 35.12°N, 89.97°W (Memphis, Tennessee area). If Hair Toffee demand is concentrated in distant regions (e.g. Pacific), the shipping distance penalty may exceed what regional averages capture.
- **Order cohort effects:** Hair Toffee has only 4 recorded orders. If these orders were placed disproportionately in early 2024 (the highest lead-time cohort), the product average would be inflated.
- **Internal production scheduling:** Hair Toffee may require specialised equipment or longer production runs at The Other Factory relative to Kazookles.
- **Capacity competition:** With only two products, any internal prioritisation of Kazookles (96 of 100 orders) may create scheduling delays for Hair Toffee.

Recommended action: A targeted investigation combining factory capacity logs, product-level shipping records, and demand geography analysis is required before any strategic decisions regarding Hair Toffee are made.

Appendix B — Computational Environment

All analysis was conducted in **Google Colab** using **Python 3.12.13**. Package versions are listed in Table 14. Random states of 42 were used throughout for reproducibility. The Haversine distance formula was implemented manually using NumPy without external geospatial libraries.

Table 14: Python Package Versions

Package	Version
Python	3.12.13
pip	24.1.2
pandas	2.2.2
numpy	2.0.2
matplotlib	3.10.0
seaborn	0.13.2
scikit-learn	(Colab default for Python 3.12)

Listing 1: Key Libraries Used

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import (
    LabelEncoder, MinMaxScaler, StandardScaler)
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import (
    RandomForestRegressor, GradientBoostingRegressor)
from sklearn.metrics import (
    mean_squared_error, mean_absolute_error, r2_score)
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
# Haversine formula implemented manually via NumPy
```